

Time-to-build Capital, Network and Business Cycle

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Abstract

We examine how the time to build for capital influences the transmission of sectoral and aggregate shocks in an economy with input-output linkages. Using granular capital project life-cycle data from India, we document substantial heterogeneity in time to build for capital across sectors. We find that sectors with stronger downstream linkages—those providing essential inputs to the broader economy and thus transmitting shocks more widely—exhibit greater capital intensity and require significantly longer time to build for capital. We incorporate this heterogeneous time to build for capital into a dynamic general equilibrium model with capital inputs and input-output linkages to analyze how these frictions shape business cycle propagation. Our results show that time to build for capital dampens short-run aggregate responses to aggregate productivity shocks while substantially increasing the persistence of their effects. Furthermore, sector-specific shocks to central sectors generate larger and more persistent aggregate effects than shocks to peripheral sectors, with this persistence being further amplified when time to build for capital is present. These findings highlight the critical role of time to build for capital in shaping both the magnitude and duration of economic fluctuations.

Keywords: Time-to-build Capital, Production Network, Business Cycle

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1 Introduction

Anecdotal and case study evidence demonstrates that investment projects require substantial construction periods. Ships, power plants, roads, and buildings all necessitate considerable time to become productive—durations that are non-negligible in macroeconomic analysis. Despite the prevalence of such lengthy construction cycles in economic activities, measures of time-to-build and their incorporation remain sparse in macroeconomic modeling. As supply chain disruptions have become increasingly pressing in recent years, a growing literature has emerged examining time-to-build and delivery lags for intermediate inputs. However, research addressing capital goods construction time remains limited.

Additionally, the time required to build these capital inputs varies dramatically across sectors. While some capital projects, like retail facilities or small-scale manufacturing equipment, can be completed within months, others necessitate significantly longer construction cycles. For example, building large-scale infrastructure such as steel mills or petrochemical refineries, typically takes three to five years from groundbreaking to operational capacity. Differences in the time required to build capital goods play a crucial role in how business cycle shocks spread through production networks. Shocks move quickly through sectors that can rapidly adjust investment and inputs, but they take much longer to filter through sectors with extended build times. At the aggregate level, this heterogeneity in time-to-build across sectors shapes the persistence of key macroeconomic variables and influences the amplitude of business cycle fluctuations.

Our work is motivated by novel evidence from a granular capital project dataset in India that allows us to track the complete life cycle of capital projects. This dataset documents substantial sectoral heterogeneity in time-to-build for capital projects, encompassing not only overall implementation duration but also delay frequency and duration. For instance, capital projects in the mining sector take an average of 9.6 years to complete, while projects in electronics, wood products, machinery, and textiles require only 2.5 years on average.

We also rank each sector by its downstream production linkage—the relative importance of its output as an input to the broader economy based on Input-Output tables—and find that sectors with strong spillover effects are also the most capital-intensive and require the longest time to build capital.

Building on these evidences, we incorporate heterogeneous time-to-build for capital into a version of the multisector real business cycle framework, based on Horvath (2000a) and Lehn and Winberry (2022). In our setup, each sector combines capital, labor, and a Cobb–Douglas bundle of intermediates from other sectors to produce gross output, with this bundle reflecting the economy’s input–output network. Following production, each sector accumulates capital by assembling investment goods from outputs of all sectors, thereby generating an investment network. In our model, capital accumulation is further subject to an exogenous time-to-build friction: investment undertaken in the current period becomes productive capital only after several periods. This delay is heterogeneous across sectors, and we calibrate sector-specific time-to-build durations using our empirical evidence that more central sectors require longer build times.

Our model highlights the role of time-to-build (TTB) capital in shaping the propagation of both aggregate and sectoral shocks through the production network. When TTB is introduced into capital accumulation, the short-run responses of value added and labor productivity to aggregate productivity shocks are muted, as capital cannot adjust immediately. However, this friction substantially increases the persistence of the shock, since investment undertaken today only augments productive capacity after several periods. At the sectoral level, productivity shocks to highly central (upstream) sectors generate larger and more persistent aggregate effects than shocks to peripheral sectors. The greater magnitude of central-sector shocks is primarily driven by the network structure, while time-to-build frictions add further persistence by delaying the response of bottleneck sectors and thereby prolonging the propagation of shocks throughout the network.

Related Literature Our work build upon and contribute to multiple strands of literature. The first uses multisector model to study sector specific shocks propagates and affect macroeconomic aggregates, for example Horvath (2000b) and other later literature build on its framework. This fast growing network literature studies how input-output networks in intermediate goods propagate idiosyncratic shocks (see Acemoglu, Akcigit, and Kerr (2016), Acemoglu, Vasco M. Carvalho, et al. (2012), Baqaee and Farhi (2019), and Bigio and La'O (2020) and more). Typically these papers feature static network models featuring linkages in the use of intermediate goods. Lehn and Winberry (2022) shows investment network is substantially different from intermediate goods network and found investment linkage matters for business cycle dynamic. Our inclusion of capital accumulation and time-to-build capital add additional channels of propagation and another way to break Hulten's lemma (Hulten 1978), potentially indicating a powerful propagation mechanism.

The second literature studies the impact of supply chain disruption within an network structure (see for example Alessandria et al. (2023), Bai et al. (2024), Vasco M Carvalho et al. (2021), Meier et al. (2020), and Schaal and Taschereau-Dumouchel (2025)). Most of this literature focuses on time spent during the delivery process, most notably international shipping delays; Some more recent work address the idea that it takes time from order to goods being shipped. Our work contribute by investigating the impact of delays of investment projects instead of intermediate goods.

The third literature we build upon and contribute to is the macroeconomic consequences of time-to-build capital. The idea of capital takes significant time to build dates back to the seminal work of Kydland and Prescott (1982); The discussion is revived and distinguished from an investment setup with adjustment cost (Lucca 2007). Fernandes and Rigato (2024) examined and confirmed the endogeneity of capital project time-to-build. We contribute to this literature by formally complementing time-to-build capital project with investment and production network.

Road Map Our paper is organized as follows. Section 2 introduces the granular Indian capital project data and documents the sectoral heterogeneity in time-to-build. We then rank industrial production sectors based on their downstream linkages in the Indian Input-Output Table and find that sectors whose inputs are more critical to their downstream sectors are more capital-intensive and also require longer construction periods for building capital. The third section presents our version of the multisector real business cycle model

with heterogeneous time-to-build capital across sectors. We investigate how time-to-build capital contributes to business cycle shock propagation and aggregate economic impact.

2 Data

We combine several data sources for the empirical analysis. Our unique micro dataset is the CapExDx database, which provides granular, project-level information on capital expenditure projects and documents key lifecycle events, allowing us to measure project completion times and delays in capital formation. To construct sector-level measures, we use the BEA Input–Output tables to calibrate production and network linkages, and the capital flow tables constructed by Lehn and Winberry 2022 to characterize the investment network across 30 nonfarm, nongovernment sectors in a given year. In addition, we draw on the BEA GDP-by-industry database to obtain annual sectoral value added and employment for the same set of sectors, following *ibid*.

CapExDx Database: We utilize data on capital investment projects from the CapExDX database, a comprehensive dataset compiled and maintained by the Center for Monitoring the Indian Economy (CMIE). CapEx provides detailed information on capital investment projects in India since 1976. A capital investment project in the database is defined as any project involving capital expenditure and capacity expansion. The database records projects with capital expenditures exceeding INR 10 million (approximately 150,000–300,000 USD, depending on the foreign exchange rate during our sample period).

The CapEx database captures key project details, including the project name, the name of the announcing company, total project cost, geographical location, industry classification, and ownership type. Crucially, it also tracks the lifecycle and status of each project, documenting milestones such as the announcement date, commencement or stalling of implementation, completion, or abandonment. This dataset allows us to measure key time-related aspects of capital projects, including overall implementation length, the duration of delays, and the frequency of delays throughout the project lifecycle. To our knowledge, it is the only granular dataset that provides such detailed information on capital project lifecycles at a wide coverage. In this paper, we use sector-specific time-to-build estimates from Indian capital projects to calibrate the time-to-build of capital in the U.S. production network at the sectoral level, under the assumption that relative time-to-build patterns across sectors are comparable between the two economies. Although it is not U.S.-based, its extensive coverage makes it highly relevant, and we believe it offers valuable insights into sectoral heterogeneity in time-to-build, with broad applicability for understanding the propagation of business cycle shocks.

BEA Input Output Table: Our research investigates how sectoral time-to-build in capital formation correlates with spillover effects and centrality in production networks, and examines how these features shape the propagation of aggregate shocks, with the goal of identifying key bottleneck industries. To measure inter-sectoral linkages and spillover effects, we utilize BEA’s input-output tables, which provides annual data from since 1940s. We employ the 2018 input-output table as our baseline to calculate each sector’s forward

linkages.

We aggregate BEA input-output private non-farm industries into 30 sectors using NAICS-based classifications, following Lehn and Winberry 2022 to facilitate calibration. Table 1 lists the sectors included in our dataset. While our input-output framework encompasses 30 sectors, we place greater emphasis on goods-producing industries—such as manufacturing, mining, and utilities—which are more capital intensive and typically involve large-scale projects with economically meaningful time-to-build. By contrast, many service sectors included in the input-output tables (e.g., trade, finance, and professional services) are substantially less capital intensive and are sparsely represented in the CapExDx project data in terms of large, long-lived investment projects. In the following section, we also document the stylized fact that time-to-build in goods-producing sectors correlates with sectoral network positions in a markedly different way than in service sectors. We therefore focus on goods - producing sectors, ensuring that our analysis is both empirically grounded and targeted toward the industries in which heterogeneity in time-to-build is most relevant for the transmission of shocks through the production network.

Table 1: The 30 Sectors Used in Our Analysis

Sector	Sector
Mining & extraction composite	Utilities
Construction	Wood products
Nonmetallic mineral products	Primary metals
Fabricated metal products	Machinery
Computer & electronic products	Electrical equipment, appliances & components
Motor vehicles, bodies, trailers & parts	Other transportation equipment
Furniture & related products	Miscellaneous manufacturing
Nondurable manufacturing (chemicals & plastics)	Wholesale trade
Retail trade	Transportation & warehousing
Information	Finance & insurance
Real estate, rental & leasing	Professional, scientific & technical services
Management of companies & enterprises	Administrative & waste management services
Educational services	Health care & social assistance
Arts, entertainment & recreation	Accommodation
Food services & drinking places	Other services (except government)

Notes: This table lists the 30 sectors used in our analysis. The sectors are aggregated from the 2018 BEA Input-Output tables and are aligned with NAICS classifications.

3 Descriptive Evidence on Time to Build

In this section, we provide motivating descriptive evidence for two key empirical patterns that underpin our research. First, we document that capital projects require substantial time to complete, with significant heterogeneity across industrial sectors in their time-to-build characteristics. Second, we demonstrate that goods-producing sectors with high network centrality (measured by downstream linkages) and large capital size are precisely those that

require more time to build projects, creating potential bottlenecks in the production network. These findings establish the empirical foundation for understanding how time-to-build frictions interact with network propagation mechanisms in business cycles.

3.1 Sector Heterogeneity in Time to Build(TTB)

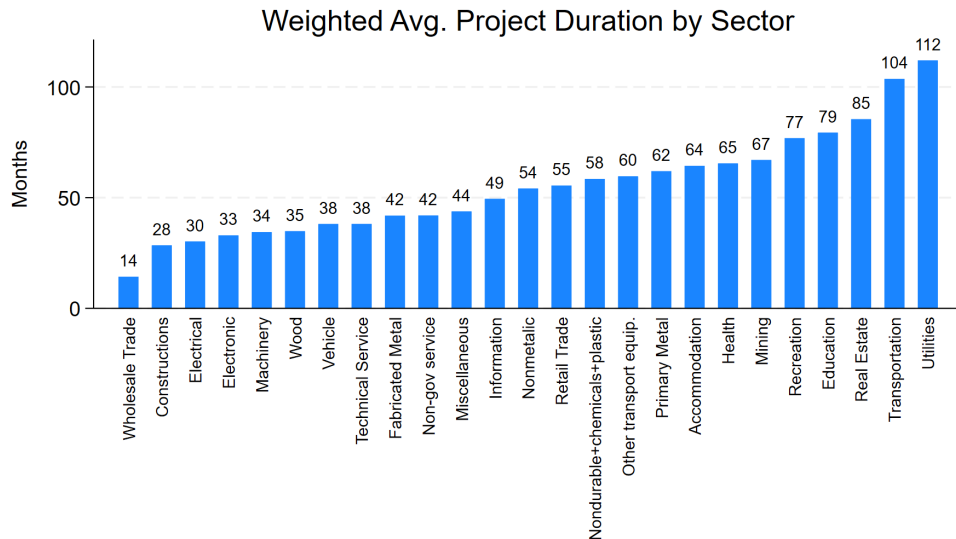


Figure 1: Average project duration by sector.

Note: Project duration is calculated as the average number of months from project initiation to completion, weighted by the cost of capital projects within each sector.

Our analysis reveals substantial heterogeneity across industrial sectors in the time required to complete capital projects. Figures 1–3 document clear evidence of sectoral heterogeneity in time-to-build across both goods-producing and service sectors. Figure 1 presents the weighted average duration of projects by sector, indicating that capital projects are inherently time-intensive endeavors with substantial cross-sectoral variation. While projects in electronics, wood, vehicles, and machinery are typically completed within 30–40 months, projects in transportation and utilities extend well beyond 100 months. Figure 2 further examines project delays, defined as the difference between actual and expected completion times. The magnitude of delays varies systematically across sectors either: projects in wood and electronic exhibit relatively modest delays of approximately one year, whereas projects in utilities, real estate and transportation experience average delays around four years. Figure 3 complements these patterns by reporting the weighted average frequency of delay events across sectors. Utilities are characterized by repeated disruptions. Together, these findings establish that capital projects require substantial time to complete and that the length, delays, and frequency of disruptions vary significantly across sectors. This heterogeneity in time-to-build underscores the importance of incorporating sector-specific frictions into analyses of investment dynamics and the propagation of macroeconomic shocks. All three

measures—project duration, delay length, and the number of delay events—are weighted by project cost to reflect the economic significance of larger projects.

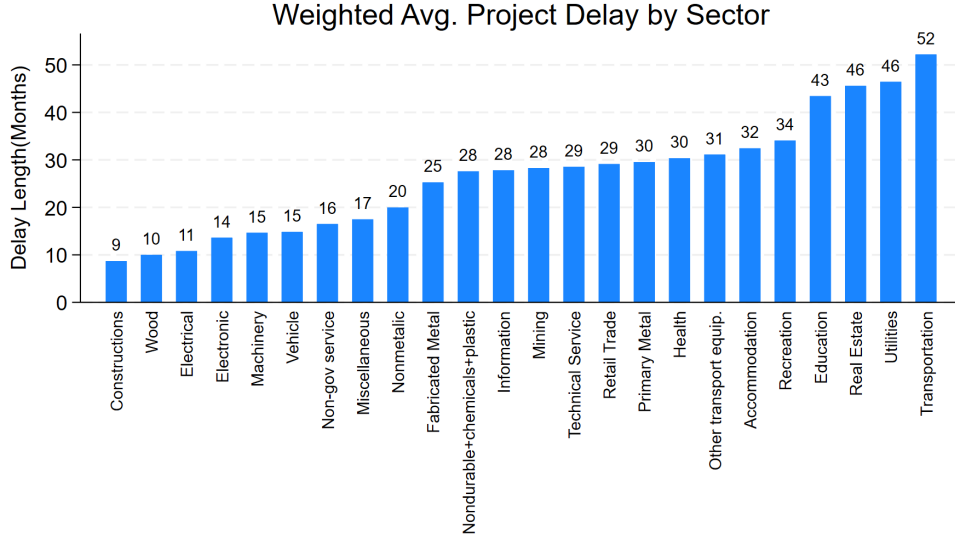


Figure 2: Average project delay length by sector.

Note: Project delay length is calculated as the difference between actual and expected completion time of the capital projects, weighted by the cost of capital projects within each sector.

3.2 TTB vs Sector Propagation Ranking

Identifying whether the sectors with longer time-to-build are also the most important for network propagation is crucial for motivating our research. If these key bottleneck sectors experience a shock and become stuck in generating the capital inputs necessary for production, their outsized importance will significantly alter shock propagation and business cycle patterns. In this part, we investigate whether there is any correlation between time-to-build duration and network centrality measures.

To quantify the relative importance of each sector, we construct sectoral rankings for the 30 US composite sectors using input–output linkage analysis, following the methodology of Acemoglu, Akcigit, and Kerr 2016. Based on the 2018 benchmark BEA input–output tables, we first transform the Make and Use tables into an industry–by–industry input–output matrix using the standard BEA methodology, so that intersectoral linkages are defined at the industry level. We then compute the *direct requirements matrix* $\mathbf{A}$, where each element a_{ij} denotes the share of sector j ’s total output sold to sector i as an intermediate input:

$$a_{ij} = \frac{\text{Sales of } j \text{ to } i}{\text{Total Sales of } j}. \quad (1)$$

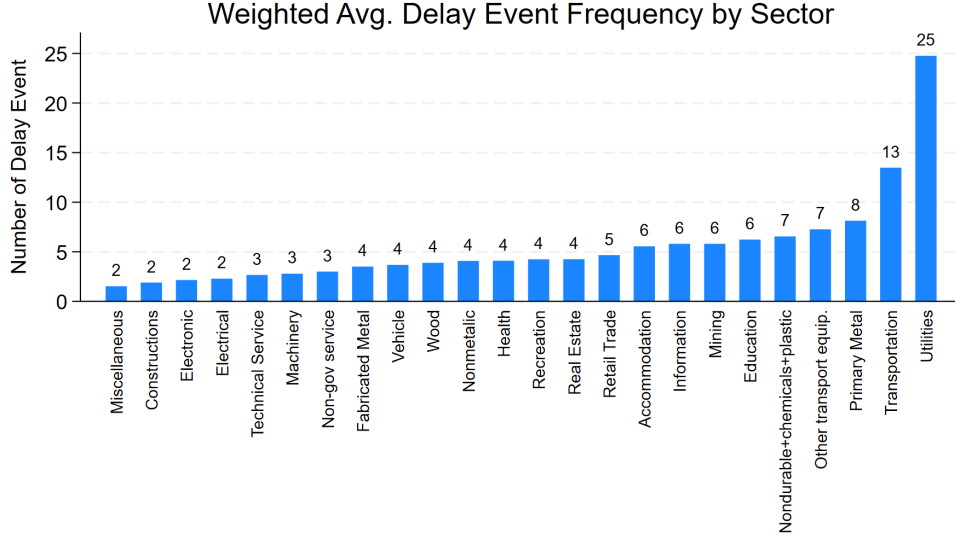


Figure 3: Average frequency of delay events by sector.

Note: Delay frequency is measured as the number of delay events reported for capital projects throughout the project life cycle in each sector, weighted by project cost.

The downstream degree of sector j is then given by the column sum

$$d_j = \sum_{i=1}^N a_{ij}, \quad (2)$$

which measures the extent to which sector j 's output is used as an input across all other sectors. Sectors are ranked in descending order of d_j , with rank 1 assigned to the sector exhibiting the strongest downstream spillovers and rank 30 to the sector with the weakest. This methodology identifies those sectors that serve as critical input providers within the US production network, where higher-ranked sectors are those whose disruptions would propagate most strongly throughout the economy.

As a robustness check, we also construct rankings based on the *total requirements matrix* $\mathbf{H}$, obtained from the Leontief inverse:

$$\mathbf{H} = (\mathbf{I} - \mathbf{A})^{-1}. \quad (3)$$

This transformation captures both direct and indirect inter-sectoral linkages by accounting for the full network of production dependencies. The total downstream importance of sector j is defined as the column sum

$$DS_j = \sum_{i=1}^N h_{ij}, \quad (4)$$

which reflects the overall economy-wide impact of a shock to sector j . In practice, the rankings obtained from $\mathbf{A}$ and $\mathbf{H}$ are very similar.

Table 2 reports the ranking of sectors by downstream degree, constructed from the direct requirements matrix $\mathbf{A}$ derived from the 2018 BEA input–output table. The ranking highlights manufacturing-related industries—such as nondurable manufacturing, primary metals, and fabricated metal products—as well as logistics-intensive sectors, including wholesale trade and transportation, which serve as key intermediaries in production and distribution. By contrast, sectors oriented primarily toward final demand, such as health care, education, and accommodation, appear at the lower end of the ranking.

Given the prominent role of utilities and real estate in standard input–output tables, their relatively low rankings in our aggregated framework may appear counterintuitive and therefore warrant clarification. Utilities rank lower in Table 2 because our analysis is based on a *30-sector aggregation* of the BEA input–output table rather than the full set of nonfarm, nongovernment industries. In the original BEA classification, utilities constitute a single industry, whereas many manufacturing and service activities are represented by multiple, more disaggregated industries. Aggregating these activities into composite sectors—such as nondurable manufacturing or transportation and warehousing—mechanically pools their downstream linkages, raising their measured centrality relative to utilities. Consistent with this interpretation, utilities rank substantially higher (around tenth) in the disaggregated 66-sector BEA input–output table.

Real estate also ranks relatively low in downstream importance in our analysis because we exclude the BEA subcategory “other real estate,” which primarily reflects leasing and capital-service flows associated with nonresidential structures. Instead, factory and building investment projects are assigned directly to the producing sectors in which the capital is used. This treatment prevents real estate from emerging as an artificial hub driven by accounting conventions and ensures that sectoral rankings reflect productive interdependencies that are relevant for capital formation and the propagation of shocks.

After constructing the sectoral rankings, we examine how sectoral time-to-build correlates with spillover effects in the production network and document a clear, positive relationship between downstream spillover strength and project implementation among goods-producing sectors. We find that goods-producing sectors with stronger spillover effects typically require longer times to complete capital investments, and they are also more capital-intensive in terms of project scale. Figure 4 shows that sectors with higher downstream spillover effects tend to have significantly longer project durations. The positive relationship between spillover rank and weighted average project duration (since a lower rank corresponds to stronger spillover effects) highlights that sectors most central to downstream sectors—such as metals, transportation, mining and utilities—take much more time to complete investment projects. These same sectors are also highly capital-intensive, as indicated by the larger bubble sizes representing total project scale of the sector. By contrast, sectors with lower spillover importance, including miscellaneous, and electrical, complete projects more quickly and on a smaller scale. Figures 5 and 6 confirm that this pattern holds across alternative measures of project time: sectors with higher spillover effects also face longer project delays and a higher frequency of delay events. Together, these results underscore that capital-intensive, high-spillover sectors are characterized by prolonged and uncertain time-to-build,

Table 2: Sectoral Downstream Importance Ranking

Rank	Sector
1	Nondurable + chemicals +plastics bundle)
2	Wholesale trade
3	Primary metals
4	Finance & insurance
5	Professional, scientific & technical services
6	Transportation & warehousing
7	Fabricated metal products
8	Administrative & waste management services
9	Management of companies & enterprises
10	Information
11	Motor vehicles, bodies, trailers & parts
12	Mining & extraction composite
13	Wood products
14	Computer & electronic products
15	Utilities
16	Machinery
17	Real estate, rental & leasing
18	Nonmetallic mineral products
19	Other transportation equipment
20	Electrical equipment, appliances & components
21	Retail trade
22	Other services (except government)
23	Construction
24	Food services & drinking places
25	Arts, entertainment & recreation
26	Miscellaneous manufacturing
27	Furniture & related products
28	Accommodation
29	Educational services
30	Health care & social assistance

Notes: Sectors are ranked by downstream degree based on the 2018 BEA input-output table. Rank 1 denotes the sector with the strongest downstream spillovers.

motivating the incorporation of TTB into production network models¹.

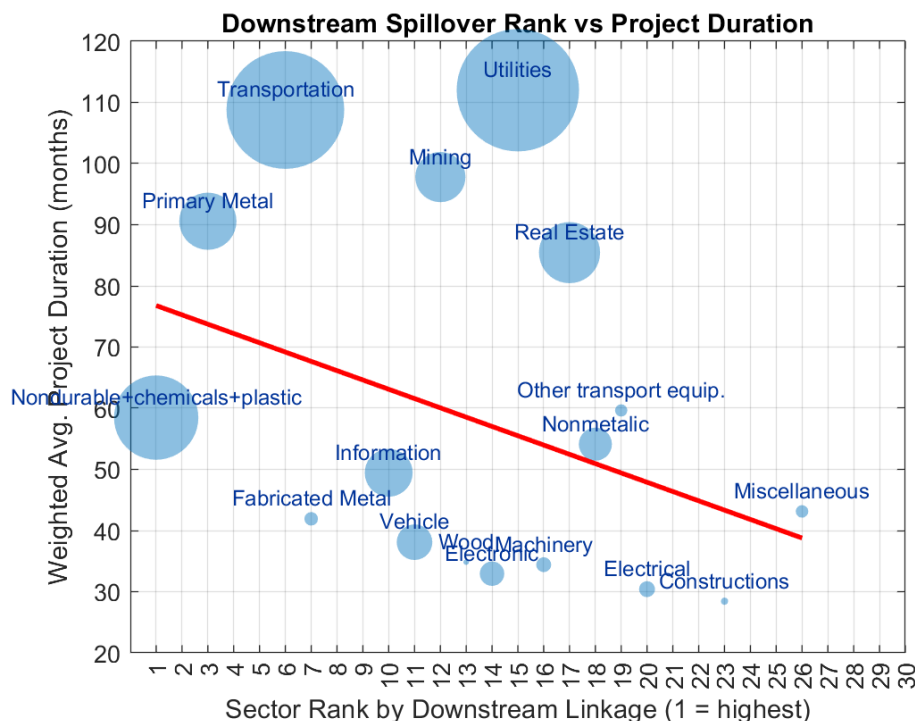


Figure 4: Relationship between downstream spillover rank and project duration.

The figure plots the weighted average project time-to-build (in months) against the sector's rank in downstream linkage importance (1 = highest). Bubble sizes are proportional to the total value of capital investment projects in each sector, as measured in the CapExDx data. The fitted red line shows the negative association between sectoral spillover importance and project duration.

One important caveat is that while we document a clear positive relationship between downstream spillover strength and sectoral time-to-build for goods-producing sectors—represented by the downward-sloping fitted line in the figure, since a lower rank denotes higher spillovers—this pattern does not extend to service sectors. In contrast, service sectors display the opposite relationship: sectors with weaker downstream spillovers tend to have longer project durations, yielding an upward-sloping fitted line, as shown in Figure 15 in the Ap-

¹Sectoral time-to-build (TTB) measures here are constructed using project-level data from the Indian CapExDx database, as comparable and comprehensive measures of TTB at the sectoral level are not available for the United States. Our approach relies on the assumption that sectoral patterns in time-to-build observed in Indian capital projects are broadly representative of, or at least comparable to, those governing capital formation in the U.S. at the sectoral level. This assumption is motivated by the technological and engineering nature of large-scale capital projects, which tend to be sector-specific rather than country-specific. We also replicate the analysis using Indian sectoral rankings constructed from the Asian Development Bank (ADB) input-output table for 16 industrial production sectors and relate these rankings to sectoral TTB measured from Indian project data. The resulting relationship exhibits a similar downward-sloping fitted line—implying a positive association between spillover strength and time-to-build (see Figure 12 - 14 in the Appendix A.1).

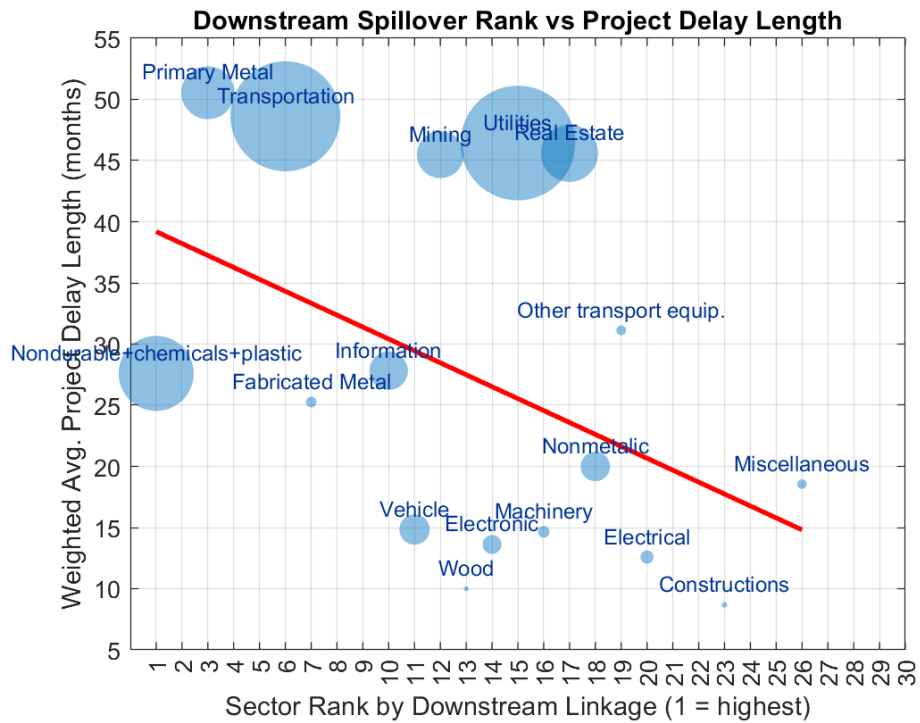


Figure 5: Relationship between downstream spillover rank and project delay length. The figure plots the weighted average project delay length (in months) against the sector's rank in downstream linkage importance (1 = highest). Bubble sizes are proportional to the total value of capital investment projects in each sector, as measured in the CapExDx data. The fitted red line shows the negative association between sectoral spillover importance and project duration.

pendix A.2. Moreover, service-sector projects are substantially smaller in scale and less capital-intensive, as indicated by their relatively small bubble sizes. For these reasons, our analysis focuses on goods-producing sectors, where time-to-build is economically meaningful and closely linked to production network spillovers.

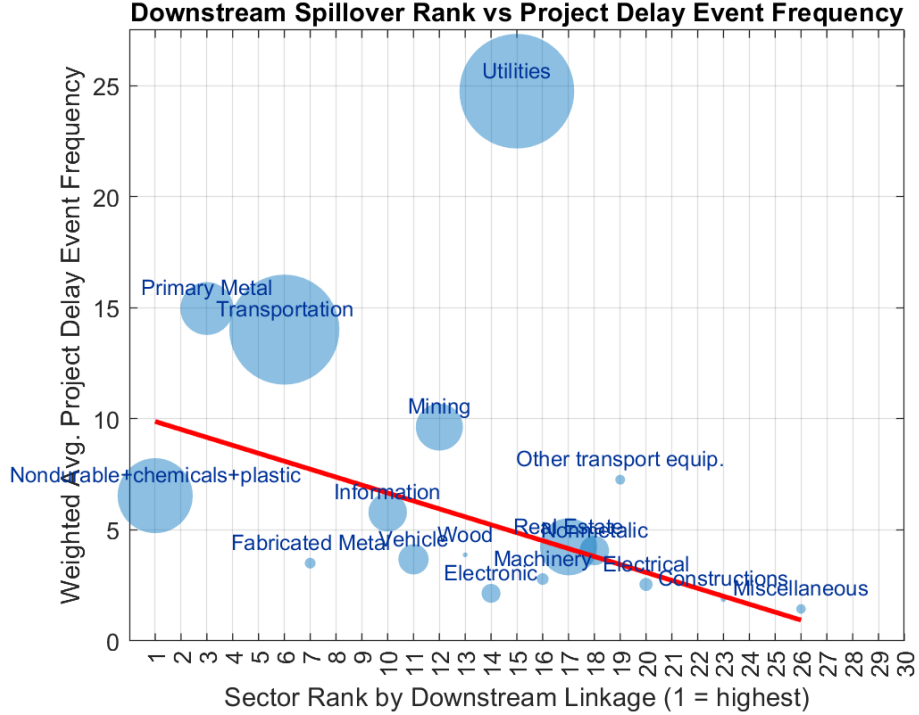


Figure 6: Relationship between downstream spillover rank and project delay event frequency. The figure plots the weighted average project delay event frequency (number of delay events) against the sector’s rank in downstream linkage importance (1 = highest). Bubble sizes are proportional to the total value of capital investment projects in each sector, as measured in the CapExDx data. The fitted red line shows the negative association between sectoral spillover importance and project duration.

4 Theoretical Model

4.1 Model Setup and Equilibrium

In this section we layout the baseline model we use to investigate the macroeconomic impact of time-to-build capital. The model economy is a multi-sector general equilibrium model populated by a representative household, N production sectors each populated by a continuum of competitive firms producing identical sector specific goods. Our baseline model is standard in the literature and is based on the Horvath (2000a) and Lehn and Winberry (2022).

Firm Firms are perfectly competitive and combine labour, capital and intermediate inputs to produce output according to the following Cobb-Douglas production function. Capital is sector specific and investment is formed using inputs from other sectors. We use subscript to denote sector $j \in \{1, 2, \dots, N\}$.

$$Y_{j,t} = A_{j,t} \left(K_{j,t}^{\alpha_j} L_{j,t}^{1-\alpha_j} \right)^{\theta_j} X_{j,t}^{1-\theta_j} \quad (5)$$

Where $Y_{j,t}$ is output in sector j at time t , $A_{j,t}$ is total factor productivity, $K_{j,t}$ is capital, $L_{j,t}$ is labour and $X_{j,t}$ is a composite of intermediate inputs. Intermediate good is formed from inputs sourced from all sectors, thus forming a roundabout production network. The composition is formed with a Cobb-Douglas function:

$$X_{j,t} = \prod_{i=1}^J X_{i,j,t}^{\nu_{i,j}}, \quad \sum_{i=1}^J \nu_{i,j} = 1 \quad (6)$$

Where $X_{i,j,t}$ is the quantity of intermediate good from sector i used in sector j at time t . Indexing is ordered such that the origin/supply sector is the first subscript and the destination/usage sector is the second subscript. Before firm chooses inputs for production, it first chooses intermediate goods bundle to minimize unit cost. This optimization yields the demand for each intermediate good and a price index:

$$P_{j,t}^X = \prod_{i=1}^N \left(\frac{P_{i,t}}{\nu_{i,j}} \right)^{\nu_{i,j}} \quad (7a)$$

$$X_{i,j,t} = \frac{\nu_{i,j} P_{j,t}^X X_{j,t}}{P_{i,t}} \quad (7b)$$

The objective of the representative firm is to choose labour, capital and intermediate goods given input prices w , r_j and P_j^X :

$$\max_{L_{j,t}, K_{j,t}, X_{j,t}} P_{j,t} Y_{j,t} - w_t L_{j,t} - r_{j,t} K_{j,t} - P_{j,t}^X X_{j,t} \quad (8)$$

Firm's optimizing behavior yields the optimal factor conditions:

$$w_t = \frac{\theta_j(1-\alpha_j)}{L_{j,t}} P_{j,t} Y_{j,t} \quad (9a)$$

$$P_{j,t}^X = \frac{(1-\theta_j)}{X_{j,t}} P_{j,t} Y_{j,t} \quad (9b)$$

$$r_{j,t} = \frac{\theta_j \alpha_j}{K_{j,t}} P_{j,t} Y_{j,t} \quad (9c)$$

We assume all the firms are owned by the representative household and thus all profits are distributed to households.

Investment After production, each sector accumulates capital by creating investment goods that combine outputs from all sectors, thereby forming an investment network. Each sector's capital accumulation is subject to an exogenous time-to-build constraint: investment made at time t will only be available as capital at time $t + \omega_j$. The dynamic of capital accumulation is thus:

$$K_{j,t+1} = (1 - \delta_j)K_{j,t} + I_{j,t-\omega_j+1} \quad (10)$$

Sector specific capital is accumulated using a sector specific investment good, $I_{j,t}$, which is produced by combining goods from all sectors using Cobb-Douglas composition:

$$I_{j,t} = \prod_i I_{i,j,t}^{\lambda_{i,j}}, \quad \sum_i \lambda_{i,j} = 1 \quad (11)$$

Where $I_{i,j,t}$ is the quantity of good from sector i used in investment good of sector j at time t . Similar to production, investors will first choose the optimal goods composition to minimize unit cost of investment. The optimization yields the demand for each good and a sector-specific investment price index:

$$P_{j,t}^I = \prod_{i=1}^N \left(\frac{P_{i,t}}{\lambda_{i,j}} \right)^{\lambda_{i,j}} \quad (12a)$$

$$I_{i,j,t} = \frac{\lambda_{i,j} P_{j,t}^I I_{j,t}}{P_{i,t}} \quad (12b)$$

Investment is performed by sector specific and perfectly competitive investors who source funds from the representative household in the form of savings and choose the optimal investment in each sectors to maximize the present value of future returns. We use these investors to conceptualize sector specific capital accumulation process. To formalize the optimization problem it is equivalent to have the representative household choose investment in each sector directly, which we will specify formally below.

Household There is a continuum of representative households in the economy that supplies labour and purchases goods from all sectors for consumption. The instantaneous utility function of the household is given by:

$$U(C_t, L_t) = \log C_t - \chi \frac{L_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} \quad (13)$$

where C_t is the household's final consumption bundle and L_t is the labour supply. χ is a preference parameter that governs the disutility of labour and $\eta > 0$ is the Frisch elasticity of labour supply. Household chooses goods from all sectors to form a final consumption bundle using a Cobb-Douglas aggregator:

$$C_t = \prod_{j=1}^J C_{j,t}^{\varepsilon_j}, \quad \sum_{j=1}^J \varepsilon_j = 1 \quad (14)$$

where ε_j is the expenditure share on good j . Similar cost minimization problem yields optimal final demand for sectoral goods and consumption price index:

$$P_t = \prod_{j=1}^N \left(\frac{P_{j,t}}{\varepsilon_j} \right)^{\varepsilon_j} \quad (15a)$$

$$C_{j,t} = \frac{\varepsilon_j P_t C_t}{P_{j,t}} \quad (15b)$$

Household then chooses the optimal consumption bundle, labour supply and investment in each sector to maximise lifetime utility:

$$\max E_0 \sum_{t=0}^{\infty} \beta^t \left(\log C_t - \chi \frac{L_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} \right) \quad (16)$$

The household faces the following budget constraint:

$$C_t + \sum_{j=1}^N P_{j,t}^I [K_{j,t+\omega_j+1} - (1 - \delta_j) K_{j,t+\omega_j}] \leq w_t L_t + \sum_{j=1}^N r_{j,t} K_{j,t} + \sum_{j=1}^N \pi_{j,t} \quad (17)$$

where total expenditure must be weakly less than the total income. Total expenditure consists of consumption and investment in each sector, and total income consists of labour income, capital income and profit income from firms. Note profit income is trivial given our firm-side assumptions. Household takes as given the wage rate w_t , rental rate of capital $r_{j,t}$ and sectoral investment price $P_{j,t}^I$ and choose consumption, labour supply and investment in each sector to maximize lifetime utility. Household's optimizing behavior yields the following conditions:

$$\frac{w_t}{C_t} = \chi L_t^{\frac{1}{\eta}} \quad (18a)$$

$$\frac{P_{j,t}^I}{C_t} = \beta^{\omega_j} E_t \left[\frac{1}{C_{t+\omega_j}} (r_{j,t+\omega_j} + (1 - \delta_j) P_{j,t+\omega_j}^I) \right], \quad \forall j \quad (18b)$$

Market Clearing and Equilibrium To complete our model, we specify the following market clearing conditions. For any sector j , its total output should either be consumed by households, used as components of intermediate goods in all other sectors and used as investment goods for all sectors. The goods market clearing conditions is thus:

$$Y_{j,t} = C_{j,t} + \sum_i I_{j,i,t} + \sum_i X_{j,i,t}, \quad \forall j \quad (19)$$

Similarly, total labour supply should be used by all sectors demanding labour, leading to the labour market clearing condition:

$$L_t = \sum_j L_{j,t} \quad (20)$$

We set the price of the final consumption good as the numeraire and let $P_t = 1$.

Equilibrium We focus our analysis on the competitive equilibrium of the economy, which we define formally below.

Definition 4.1. A competitive equilibrium is a set of prices $\{P_{j,t}, P_{j,t}^I, P_{j,t}^X, w_t, r_{j,t}\}_{j=1}^N$ and allocations $\{C_t, L_t, C_{j,t}, L_{j,t}, K_{j,t}, I_{j,t}, Y_{j,t}, X_{j,t}, I_{i,j,t}, X_{i,j,t}\}_{j=1}^N$ such that:

- Current state is defined by $\{K_{j,t}\}$ and $\{I_{j,t-\omega_j+1}, I_{j,t-\omega_j+2}, \dots, I_{j,t-1}\}$ for all j ;
- Given prices and state, the representative household chooses $\{C_t, L_t, C_{j,t}, L_{j,t}, I_{j,t}\}_{j=1}^N$ to maximize lifetime utility subject to the budget constraint;
- Given prices and state, each firm in sector j chooses $\{L_{j,t}, K_{j,t}, X_{j,t}\}$ to maximize profit subject to the production function;
- Given prices and state, investors in sector j choose $\{I_{j,t}\}$ to maximize the present value of future returns;
- Prices clears goods and labour markets;
- The price of the final consumption good is set as the numeraire, $P_t = 1$.

We then solve the model numerically and with log-linearization around the deterministic steady state. Our first order approximation of the model allow us easily compute impulse responses to various shocks of interest even with the presence of large number of state variables from both our time-to-build specification and network structure. We are cautious that our baseline linearized model abstracts from potentially important non-linear responses. In our future work we will check the robustness of our results using non-linear solution methods.

5 Quantitative Exercise

In this section we provide a quantitative exercise to illustrate the implications of our time-to-build specification. The goal of this exercise is to illustrate how time-to-build capital can affect the transmission of business cycle shocks in a multi-sector economy comparing to standard specification. We perform three different exercises: 1. Examine the impulse responses to an aggregate TFP shock and the mechanism behind the differences in propagation and amplification comparing to No TTB and homogeneous TTB specifications. 2. Examine the impulse responses to sectoral shocks at different parts of the network; 3. Utilize the model to evaluate bottleneck sector in transmitting aggregate productivity shock.

5.1 Calibration

We calibrate the structural parameters of the model using values from the existing literature, US national account and our sectoral time-to-build measures. In future work, we plan to estimate these parameters by matching model steady state to key empirical moments. The model features $N = 30$ sectors, which are mapped directly to the sectors used in our stylized facts. Accordingly, we rely on the BEA input-output tables to discipline the production-side parameters. The model is currently calibrated to data from 2018.

Taking the value from Lehn and Winberry 2022 for 2018, the share of primary inputs in production, θ_j , is calibrated as the ratio of sector j 's value added to gross output. The labor share, $1 - \alpha_j$, is measured as average labor compensation as a fraction of total value added. The intermediate input-output coefficients, ν_{ij} , are given by sector j 's expenditure on intermediate inputs from sector i as a share of its total intermediate expenditure. The investment network parameters, λ_{ij} , are calibrated as the share of sector j 's total investment expenditure purchased from sector i . The structure of production network is shown in figure ??, and that of the investment network is shown in figure ?. Sector-specific depreciation rates, δ_j , are set to the implied depreciation rates in the data. Consumption shares, ε_j , are calibrated as the average share of total consumption expenditure allocated to sector j 's output. Finally, we set the discount factor to $\beta = 0.9$, normalize the disutility-of-labor parameter to $\chi = 1$, and set the Frisch elasticity to $\eta = 2$.

Table 3: Parameter Value for Quantitative Exercise

Parameter Name	Description	Value
β	Discount factor	0.9
$1/\eta$	Inverse Frisch elasticity	0.5
χ	Labor disutility weight	1
ρ	Persistence of Productivity Shock	0.7
δ	Depreciation rate	0.1
α	Capital share	BEA National Account
θ	Value-added share	BEA National Account
λ_{ij}	Investment network weight	Lehn and Winberry 2022
ν_{ij}	Production network weight	BEA IO Table

For the time-to-build (TTB) calibration, we normalize the sector with the shortest construction lag to one period and calibrate sector-specific TTB parameters using project-level data. For comparison, we consider a model without time-to-build frictions, in which all sectors face a uniform one-period construction lag ($\omega_j = 1$). We refer to the model with sectoral TTB as the baseline model and the model without TTB as the standard model. We also consider a model with homogeneous TTB, in which all sectors face the same construction lag equal to the economy-wide average TTB. This allows us to compare the effects of allowing for realistic sectoral heterogeneity in TTB versus a uniform lag structure used in prior literature.

5.2 Response to Aggregate Shock

We start with the impulse responses to an aggregate TFP shock that hits all sectors equally. Figure 7 illustrates the impulse responses of key macroeconomic variables of model with heterogeneous time-to-build (TTB) and no-TTB. Of the aggregate variables, we focus on aggregate value added/GDP, aggregate investment, aggregate consumption, and labour productivity. Comparison between the impulse-responses shows that by adding time-to-build significantly alters the propagation and amplification of aggregate TFP shock comparing to the standard model.

Immediately by observing the on-shock patterns of impulse responses, we find similar amplification or stronger amplification for models with TTB comparing to the standard model. For

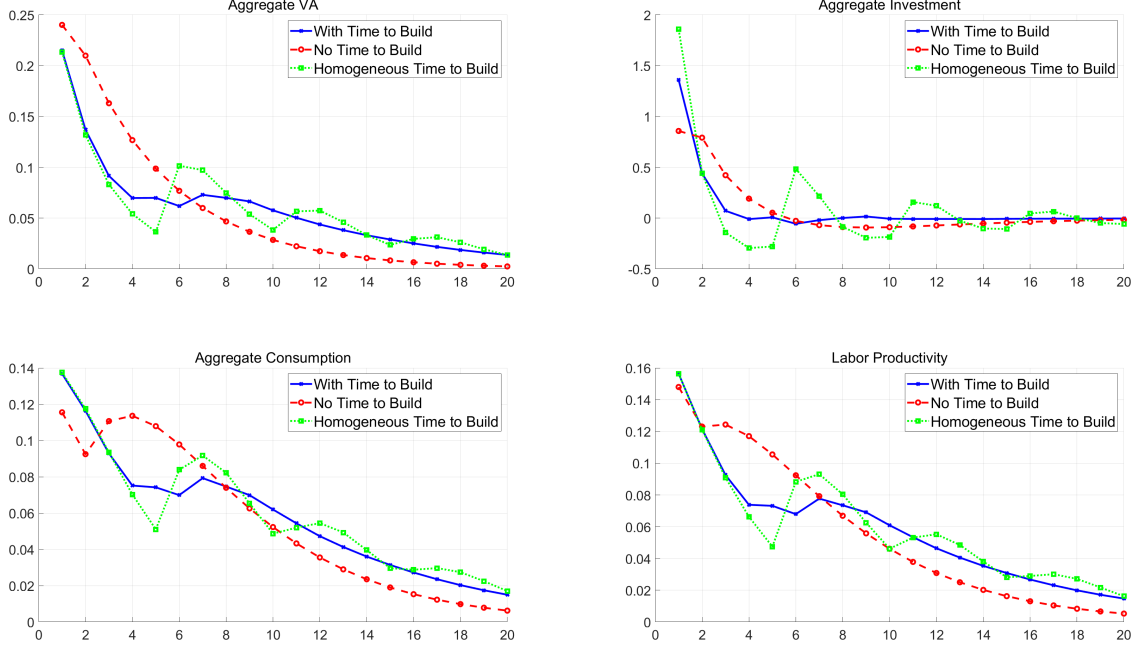


Figure 7: Impulse response function to aggregate productivity shock. The plots show the impulse response of (top left to bottom right) aggregate value added, aggregate investment, aggregate consumption and labor productivity to a positive aggregate productivity shock. With time-to-build uses heterogeneous TTB derived from data, no TTB sets $\tau_j = 1, \forall j$, homogenous TTB sets $\tau_j = \bar{\tau}, \forall j$.

aggregate value added and labour productivity the on-shock response is slightly muted comparing to a standard model, yet the effect persists for a longer period of time. Consumption is slightly more responsive on impact. Investment is significantly amplified comparing to the standard model, with on-impact responses almost doubled. In terms of persistence, adding time-to-build creates two main distinctions: faster decay immediately after the shock and stronger persistence in the medium term onwards. This pattern holds for both homogeneous and heterogeneous TTB/baseline specification, and is observed for all aggregate variables of interest. When comparing our heterogeneous TTB specification to the homogeneous TTB specification, we observe that the homogeneous TTB model exhibits significant oscillatory responses: For investment, there is a significant drop below steady state after two period of shock, then investment jumps up to another peak around period 5 and 6. Though not as significant, we observe this secondary peak (and to even lesser extend the third peak) in value added and consumption as well.

TTB plays a vital role in generate these response patterns. The intuition for the amplification of investment's response is that with a high time-to-build lag, firms find it optimal to invest more in the immediate period both because the productivity is the highest then, and because when the investment is completed it will generate higher output and return comparing to later investments. However, after the first few periods, the investment made is ample in the pipeline such that when the investment made today becomes productive, the marginal return to capital drops so that investing is no longer profitable even when produc-

tivity shock still persists. This forms the trough in investment after the pick. When the investment in the pipeline clears, firms find it optimal again to invest again and repeat the investment cycle. The oscillation is then an artifact of this investment cycle induced by time-to-build lag. Such pattern is also observed in time-to-build structure in intermediate goods in Schaal and Taschereau-Dumouchel 2025. When the economy is subjected to the same investment cycle, oscillation manifest also on aggregate variables. Adding heterogeneity across sector allow different sectors to have their own investment cycle and thus smooth out the oscillation at the aggregate level and create more realistic impulse responses, comparing to the homogeneous TTB specification.

Another direct result of this investment pattern and time-to-build lag is the prolonged persistence and the second smaller hump: when the investment in the pipeline eventually becomes productive, it allow firms to produce more output even when the productivity shock has dissipated. Thus, the second half of the impulse response is dominated by increase production/aggregate consumption even when investment has returned to steady state.

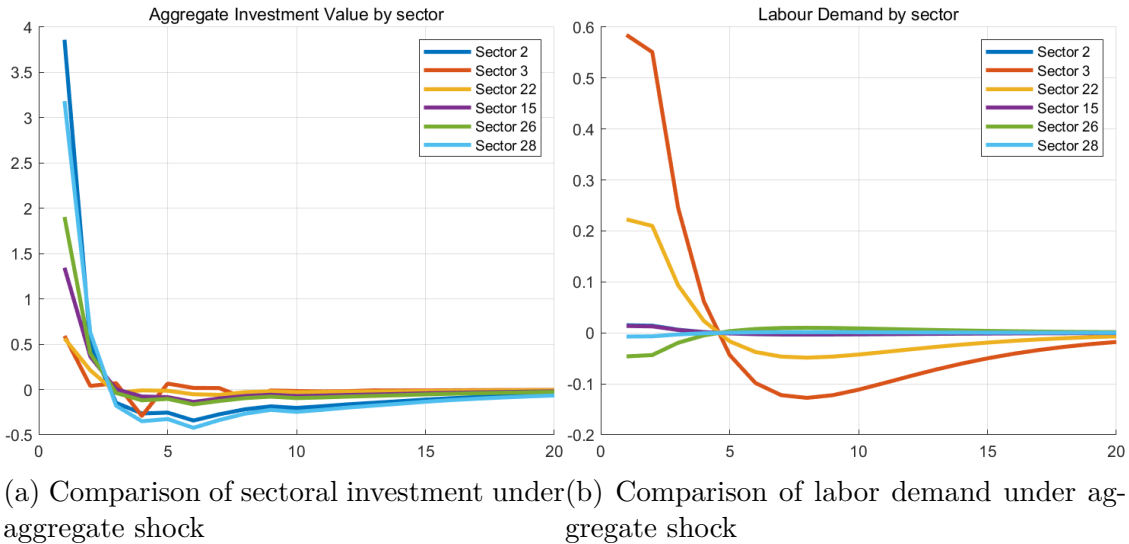


Figure 8: Sectoral responses to aggregate productivity shock: sectoral investment (left) and sectoral labor demand (right); Sectors are Utility (2), Construction (3), Technical service (22), Manufacturing cluster (15), Health (26), Accommodation (28)

We then explore the sectoral responses to analyse how shocks are amplified and propagated through the network structure. We focus on 6 sectors that have distinct production and network locations. These sectors are: 1. Investment hubs²: Construction (3) and Technical service (22); 2. Manufacturing cluster (15) as upstream sector in production network and health (26) as downstream sector; 3. High TTB sectors: Utilities (2) and accommodation (28). Figure 8a presents the impulse responses of sectoral investment of said sectors and figure 8b the impulse responses of sectoral labour demand. From the pattern in the first figure, we observe that investment is significantly amplified for the sectors with high time-to-build lag, and both the manufacturing cluster and health sector. On the other hand,

²Investment hubs are sectors that heavily used in investment goods regardless sector. In our specification they are Construction, Machinery and Professional services. A visualization can be found in figure ??.

investment responses of construction and technical service sectors are minimal. Conversely, in figure 8b we observe that construction and technical service sectors experience the highest increase in labour demand, while the other sectors exhibit more muted responses.

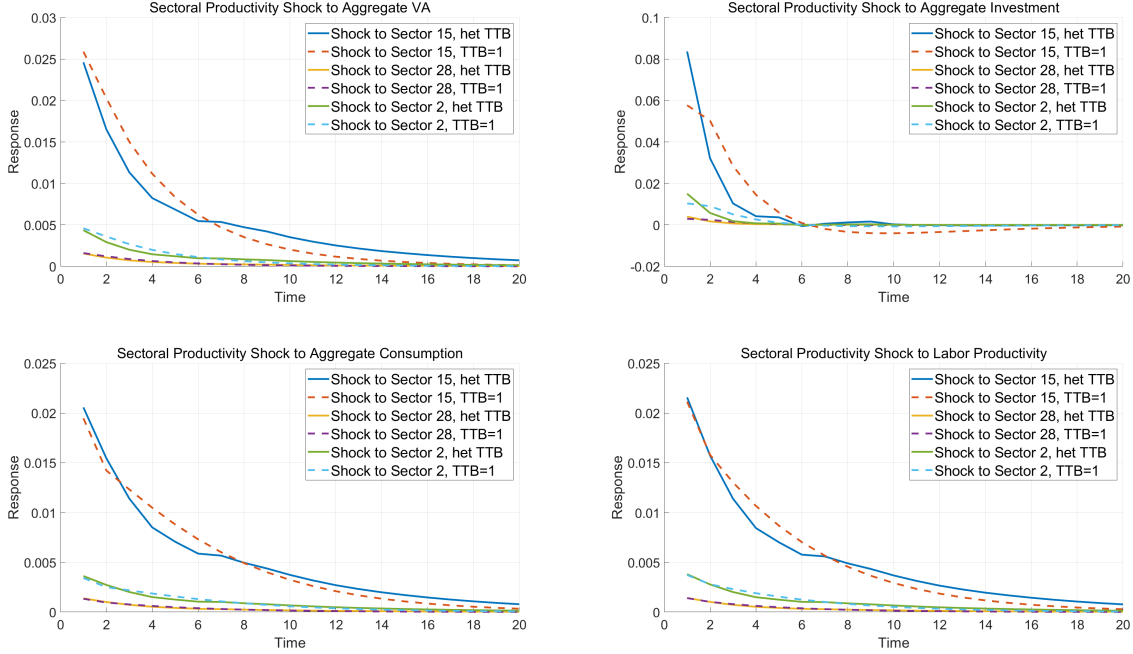


Figure 9: Impulse response functions to sectoral productivity shock. The plots show the impulse response of (top left to bottom right) aggregate value added, aggregate investment, aggregate consumption and labor productivity to a positive sectoral productivity shock. Sectors shocked are Utilities(2), Nondurable+chemicals+plastic manufacturing(15) and Accommodation(28); Comparison are done between heterogeneous TTB and no TTB.

The reasons for the induced high investment vary: for utility and accommodation sectors, the high time-to-build lag means more investment is needed to fill the pipeline and thus investment response is amplified; For manufacturing cluster, because it is located at the top of the production network and supplies all other sectors with intermediate goods, the aggregate productivity shock generates induced additional demand through production network by going upstream; For the health sector, though it is located downstream, its high consumption share in the good basket means that the induced increase in consumption generate higher demand and thus increases investment.

As the demand for investment goods is amplified for these sectors, investment hubs increases their labour demand to meet such demand, hence the great amplification on shock. Once the initial surge in investment dwindles and the decline overshoots the steady state level. Conversely, the initial increase congests labour market and prevents sectors from fully respond to the productivity shock, which explains the muted labour demand responses for these sectors. In all, time-to-build changes the investment dynamic significantly and allow for stronger propagation after an initial faster decay.

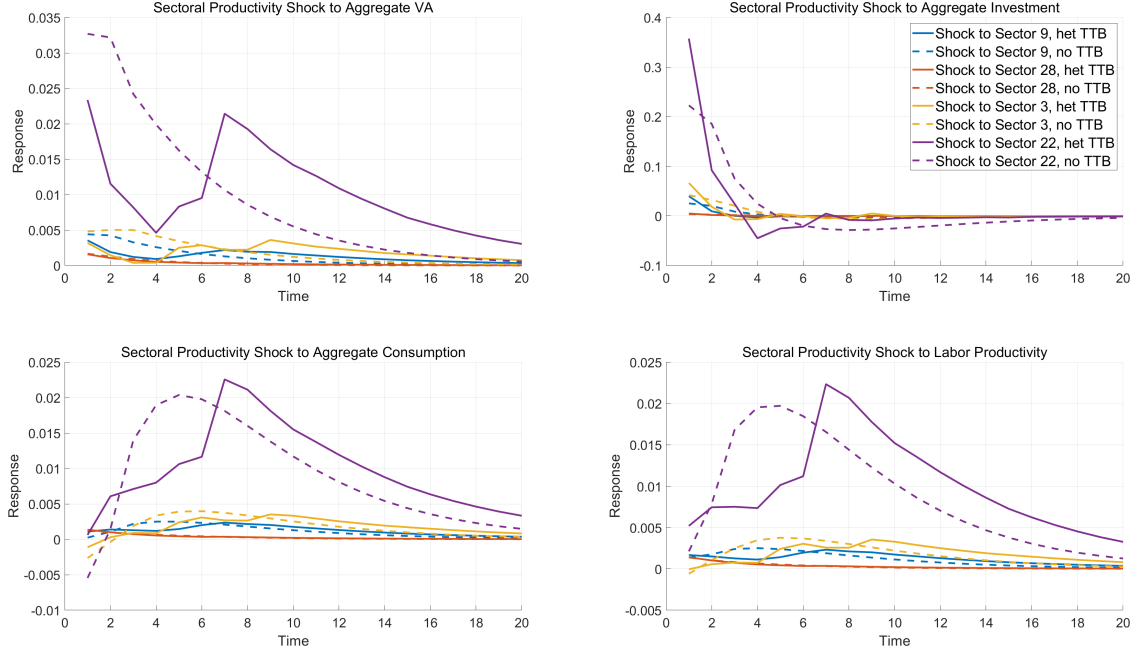


Figure 10: Impulse response to sectoral shock to 1. Non investment hubs: Electronic (9) and Accommodation (28) and 2. Investment hubs: Construction (3) and technical services (22).

5.3 Response to Sectoral Shock

Next, we apply our parameterized model to investigate the response to sectoral shocks. We first consider the importance of network location and TTB in the impulse responses to sectoral shocks. Figure 9 shows aggregate impulse responses to shocks in three sectors: Manufacturing cluster (15), Accommodation (28) and Utilities (2). The first two sectors capture location differences within production network and utility represents sector with heavy TTB. We also considered comparison between heterogeneous TTB and no TTB.

The results is intuitive: shocks to upstream sectors (manufacturing cluster) are significantly amplified and propagated through our network structure, while shocks to the periphery (accommodation) are much less so. Our time-to-build structure generate similar alterations to the shape of impulse responses as in the aggregate shock case. The magnitude of which also depends on the amplification of network. Shocks to sectors with high TTB generate moderately amplified responses, indicating that for sectoral shock propagation network location plays a more important role.

We now explore how sectoral shocks get amplified and propagated within the investment network. To do that, we consider shocks to investment hubs and non-investment hubs. For investment hubs, we consider shocks to construction (2) and professional services (22); for non-investment hubs, we consider shocks to electronic (9) and accommodation (28). Figure 10 presents the impulse responses of key aggregate variables to these shocks. We also include comparison between parameterization with heterogeneous TTB and no TTB.

Comparison across sectors for heterogeneous TTB model shows a clear distinction between

investment hubs and non-investment hubs: shocks to investment hubs are significantly amplified and propagated through the investment network, particularly for professional service sector. On the other hand, shocks to non-investment hubs generate minimal aggregate responses. Comparing between shocks to investment hub, on-shock consumption is none responsive, even negative. This is because shocks to investment hubs has a much stronger effect in lowering the cost of investment and thus inducing higher investment, over any effect directly on consumption. Additional negative effect induced on consumption is through congested labour market reducing final goods production as investment is burgeoning. As the economy go through the investment cycle, a second peak emerges as the investment in the pipeline matures and become productive. For professional service sector, this effect is even large enough to generate a similar aggregate VA increase comparing to the on-shock response.

The standard model (no TTB), we observe a similar hump shape response in consumption due to the intuition above. Adding TTB postpone the arrival of the peak and add further persistence. For aggregate VA, the standard model generates larger on-shock response, likely due to smaller congestion effect from smaller investment surge. TTB seems to be important in generating the two peak pattern for investment-hub-shock. The additional persistence in the longer horizon is amplified when the shock originates from investment hubs. In all, shocks to investment hubs create richer interplay between network and time-to-build structure.

5.4 Bottleneck Sector

The last important quantitative exercise we do with our parameterized model is to identify bottleneck sectors in generate the differences in aggregate responses to shocks. For this particular exercise, We measure how much a sector is a bottleneck by how much its sectoral TTB changes the shape of aggregate VA impulse response to an aggregate productivity shock. If a sector is the top bottleneck sector, when changing its TTB to 1 (no TTB), the aggregate VA impulse response resulting from the new parameterized model will deviate the most from the baseline model's impulse response.

There are multiple ways to measure how different two impulse responses are. In this exercise we consider two measures: 1. Kurtosis of the impulse response; 2. Distance ratio between two impulse responses. For kurtosis, we assume the impulse response is a probability distribution and calculate its kurtosis using moment information. For distance ratio, we simply measure the relative difference in the two impulse responses to the baseline model at each period and sum them up. We present the result from the first measure in the main text, but results using distance are similar.

Figure 11 presents the top 10 bottleneck sectors identified using kurtosis measure. Sector ordering using distance measure is similar, as shown in figure 18. We find real estate is the top bottleneck sector, and by far the most important one. Other sectors are utilities, manufacturing clusters (i.e. nondurable goods, chemical, plastic) and more. The mix of bottleneck sectors indicates that both network location and time-to-build matters: Utilities are importance as it has the highest TTB, while manufacturing clusters are important as they are located upstream in the production network. Real estate likely combines both

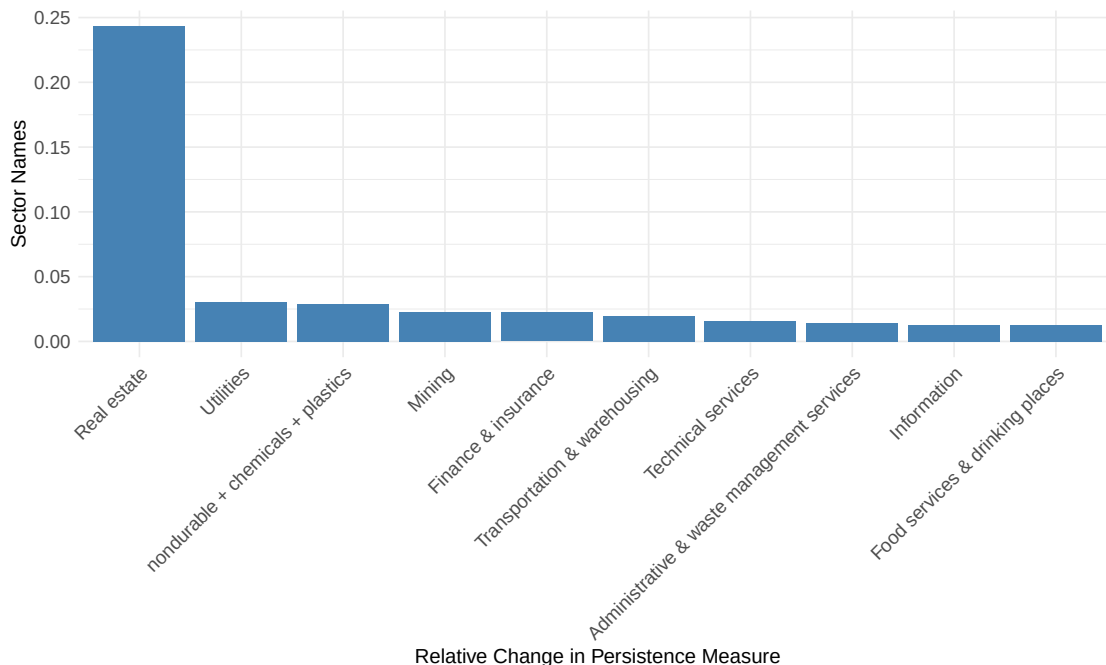


Figure 11: Identified top 10 bottleneck sectors from time-to-build, measured using kurtosis.

features.

A distinct feature of the identified bottleneck sectors is that real estate alone can generate a significant portion of the observed differences in impulse responses. This is because of our calibration, real estate’s relative high TTB and its location in the production network. Real estate is heavily capital intensive, and value added intensive, while reasonably supplying intermediate goods to all other sectors. Thus, when its TTB changes, its high capital intensity means that investment response is highly amplified and also it is not effected by the congestion effect from labour market. The high value added intensity means it is less affected by upstream sectors, while it will also respond to downstream sector demand changes. In all, production network location and calibrated TTB matters in deciding whether a sector is a bottleneck or not, while other parameters are important in determining how much a sector alters the aggregate response.

6 Conclusion

This paper examines how heterogeneous time-to-build (TTB) for capital shapes the propagation of both aggregate and sector-specific shocks through production networks. Using granular project-level data from India, we provide novel evidence of substantial heterogeneity in TTB across sectors. Moreover, we show that sectors requiring longer build times are also those that are central in the input–output network, making them bottlenecks or hubs with disproportionate influence on aggregate activity and shock transmission. We then incorporate heterogeneous TTB into a dynamic multi-sector general equilibrium model with both production and investment networks. The model demonstrates that TTB generate similar

on-shock aggregate response to productivity shocks, faster decay but substantially increases their persistence. Time-to-build structure also introduce an investment cycle pattern that features bunching investment at the start and creating oscillation in the response to shock. Allowing for heterogeneous TTB smooths out the oscillation. This front loading mechanism and TTB together explains the amplification, fast initial decay and strong mid-term persistence, and the separation of production and investment is amplified by congestion in labour market.

Under sector-specific shocks, our heterogeneous TTB model with networks produce similar results comparing to standard network models while preserving the same pattern change as the responses to aggregate shock. Shocks to upstream sectors generate larger responses than more downstream sectors. Responses are highly non-monotonic when shocks originated in investment-hubs and yield a two-peak pattern. We also find under current parameterization real estate is the most important bottleneck sector that drives the shape of the impulse response from standard no-TTB model. The importance of a sector in driving the difference depends both on calibrated TTB and its network location.

Future research will extend the analysis in three directions. First, we will incorporate capital accumulation in the form of purchasing machinery and equipment to match the actual capital accumulation process. Second, although our current focus is on productivity shocks, the propagation mechanism is not specific to this shock type, and future work will study the transmission of alternative disturbances. Third, while we treat sectoral TTB as exogenous in the present model, preliminary evidence suggests that TTB is itself responsive to business cycle conditions. An important extension will therefore be to endogenize TTB by allowing for delays and speeding up and analyze how it interacts with shocks in shaping aggregate dynamics.

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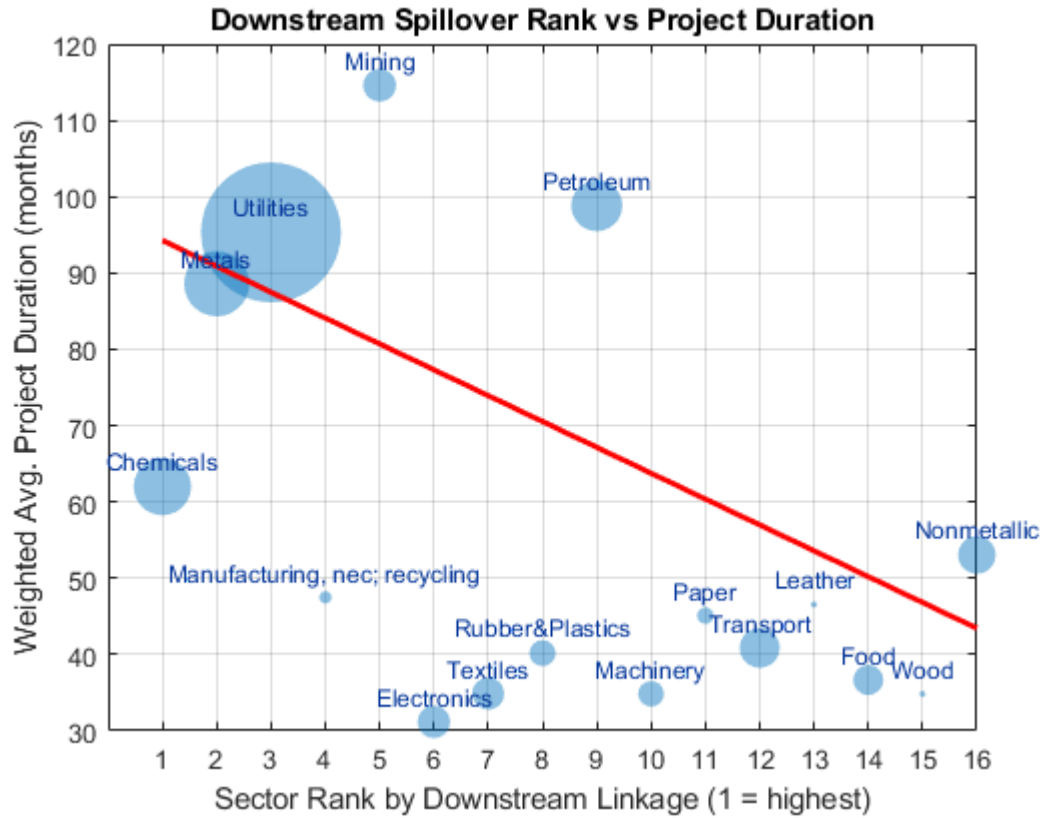


Figure 12: Relationship between downstream spillover rank and project duration. The figure plots the weighted average project time-to-build (in months) against the sector's rank in downstream linkage importance (1 = highest). Bubble sizes are proportional to the total project cost of each sector. The fitted red line shows the negative association between sectoral spillover importance and project duration.

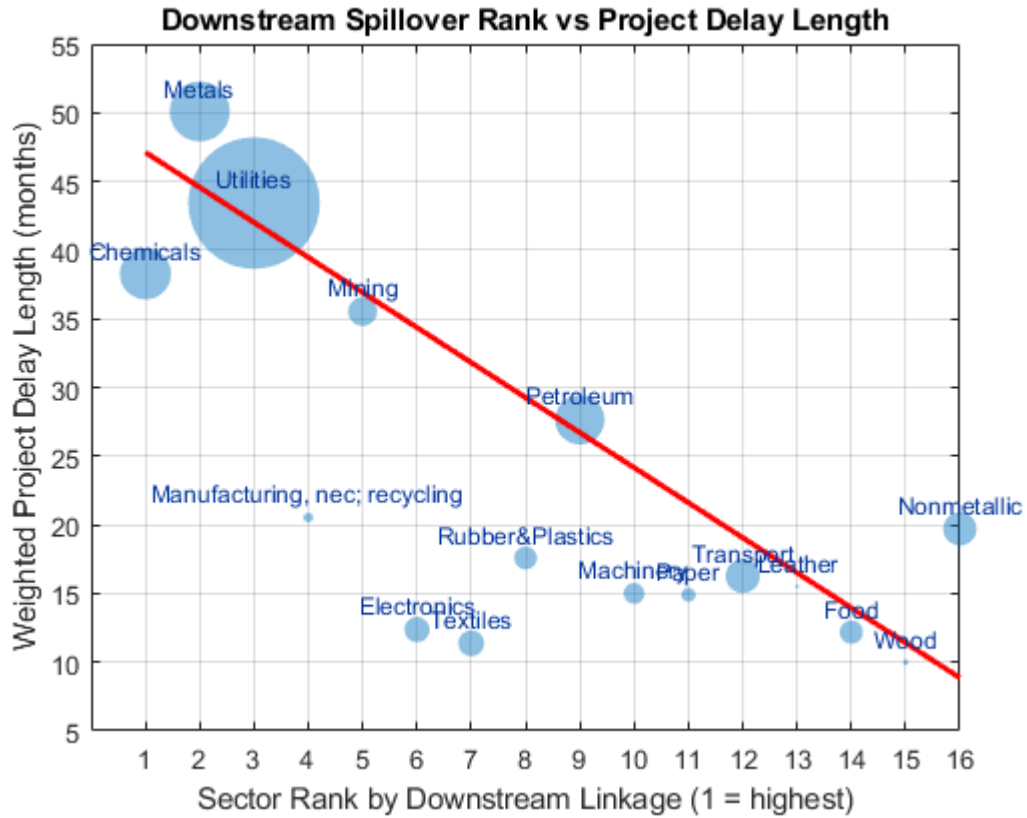


Figure 13: Relationship between downstream spillover rank and project delay length. The figure plots the weighted average project delay length (in months) against the sector's rank in downstream linkage importance (1 = highest). Bubble sizes are proportional to the total project cost of each sector. The fitted red line shows the negative association between sectoral spillover importance and project duration.

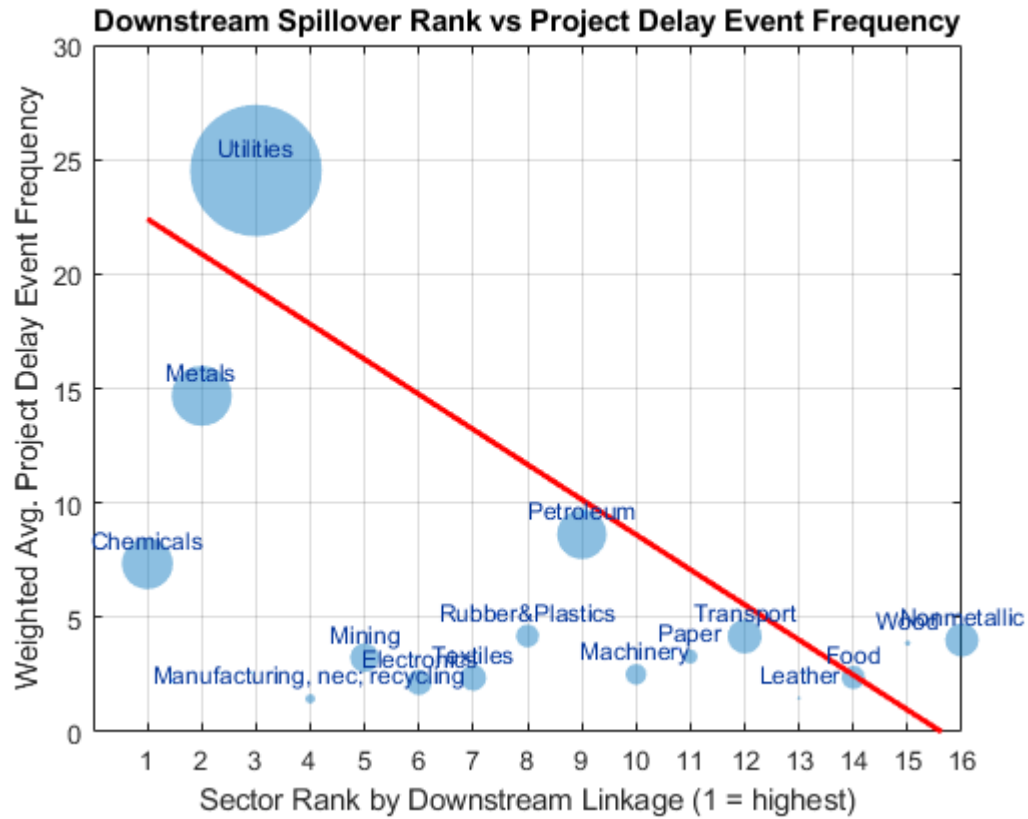


Figure 14: Relationship between downstream spillover rank and project delay event frequency.

The figure plots the weighted average project delay event frequency (number of delay events) against the sector's rank in downstream linkage importance (1 = highest). Bubble sizes are proportional to the total project cost within each sector. The fitted red line shows the negative association between sectoral spillover importance and project duration.

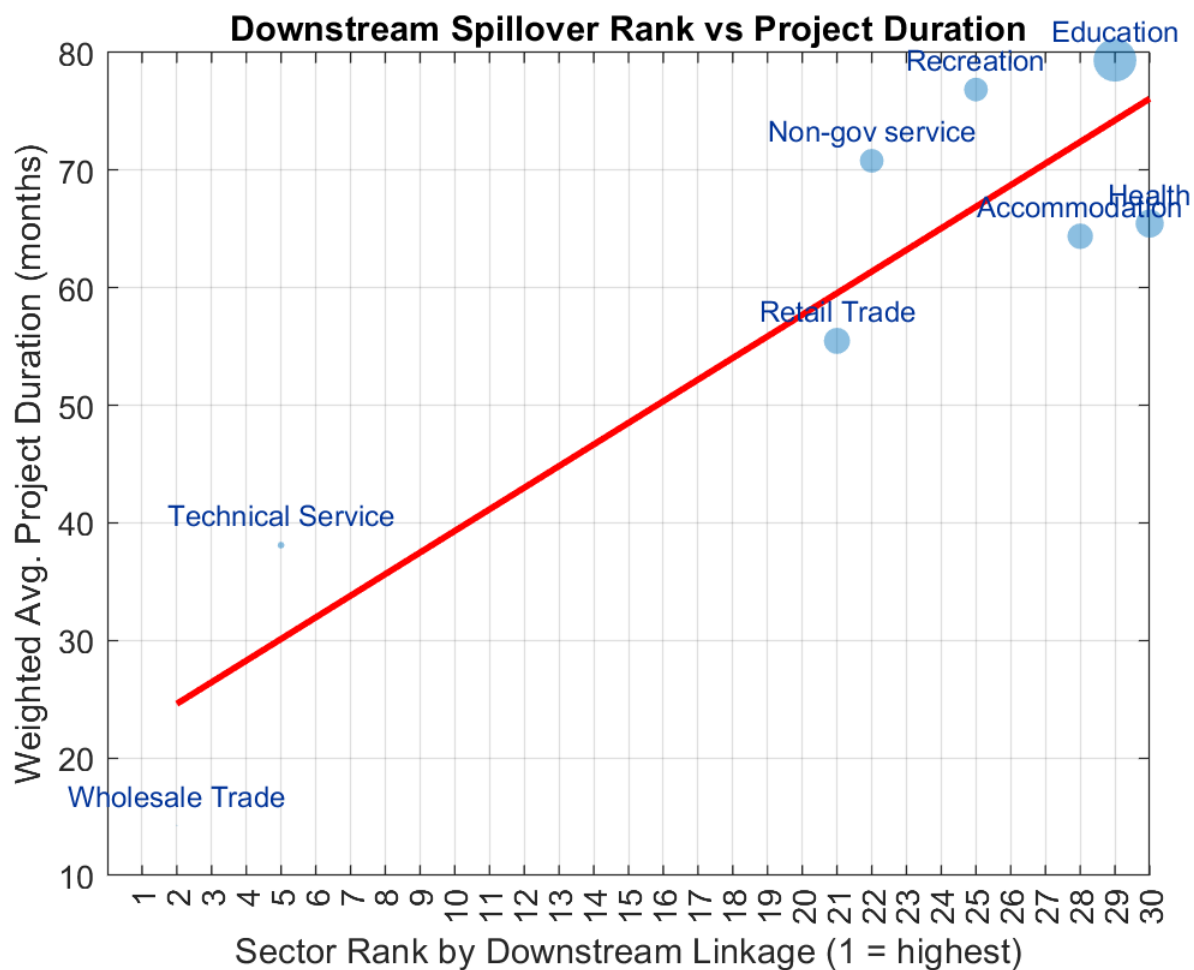


Figure 15: Caption

A Appendix

A.1 TTB vs Sector Propagation Ranking - Indian IO Table

A.2 TTB vs Sector Propagation Ranking - Service Sector

A.3 Calibration Network Structure

A.4 Additional Quantitative results

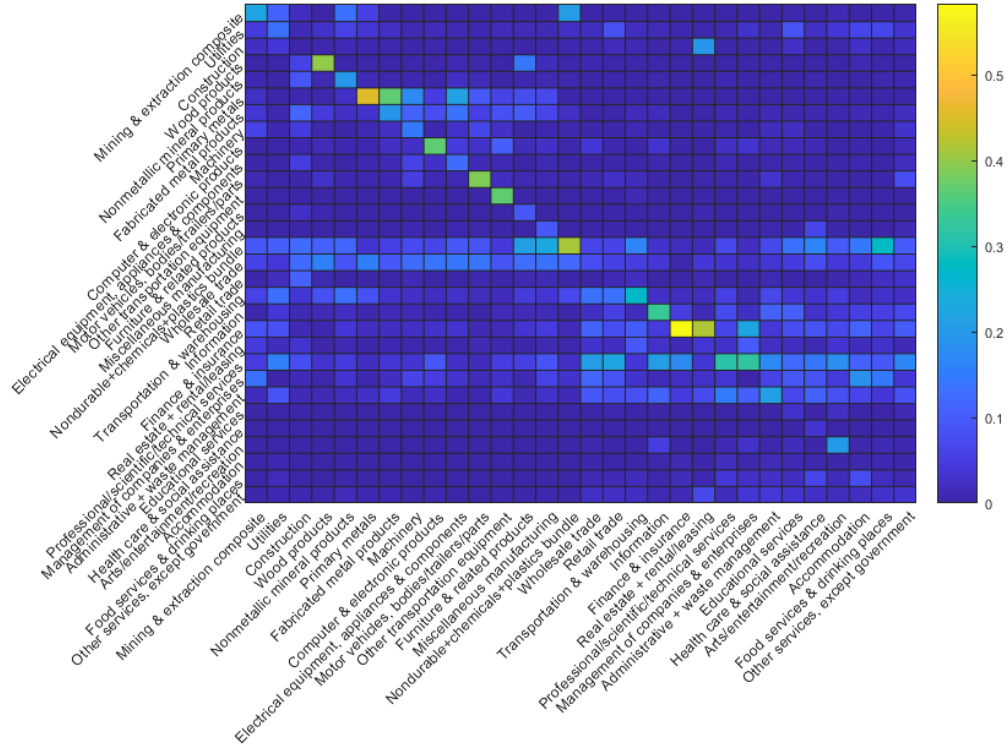


Figure 16: Input-output matrix heatmap used in model parameterization.

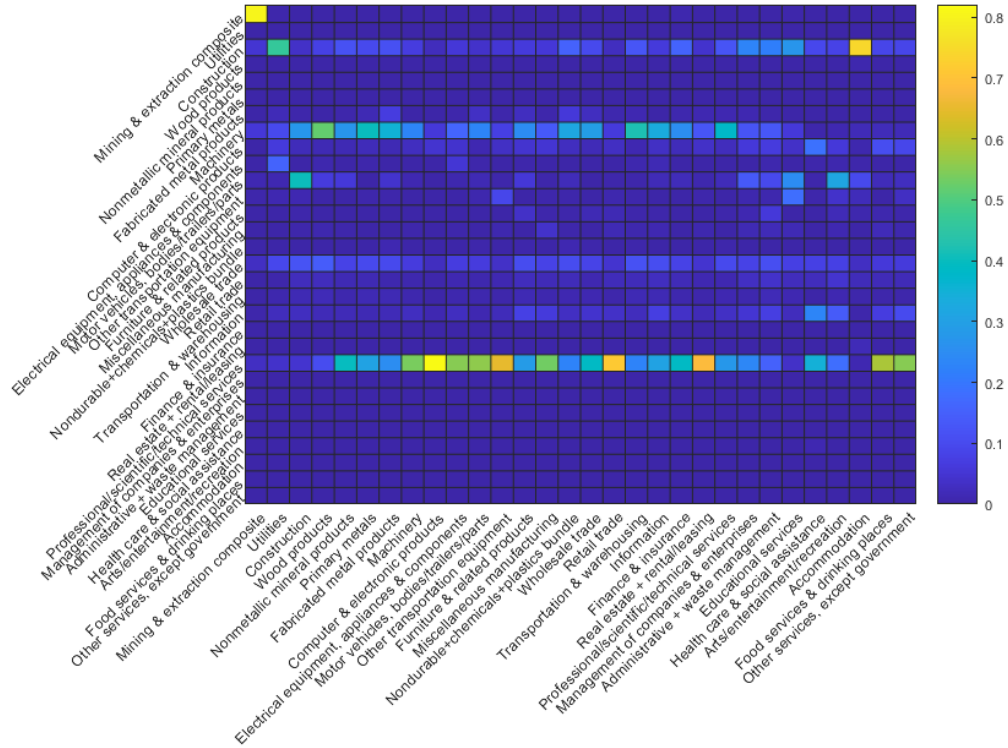


Figure 17: Investment network matrix heatmap used in model parameterization.

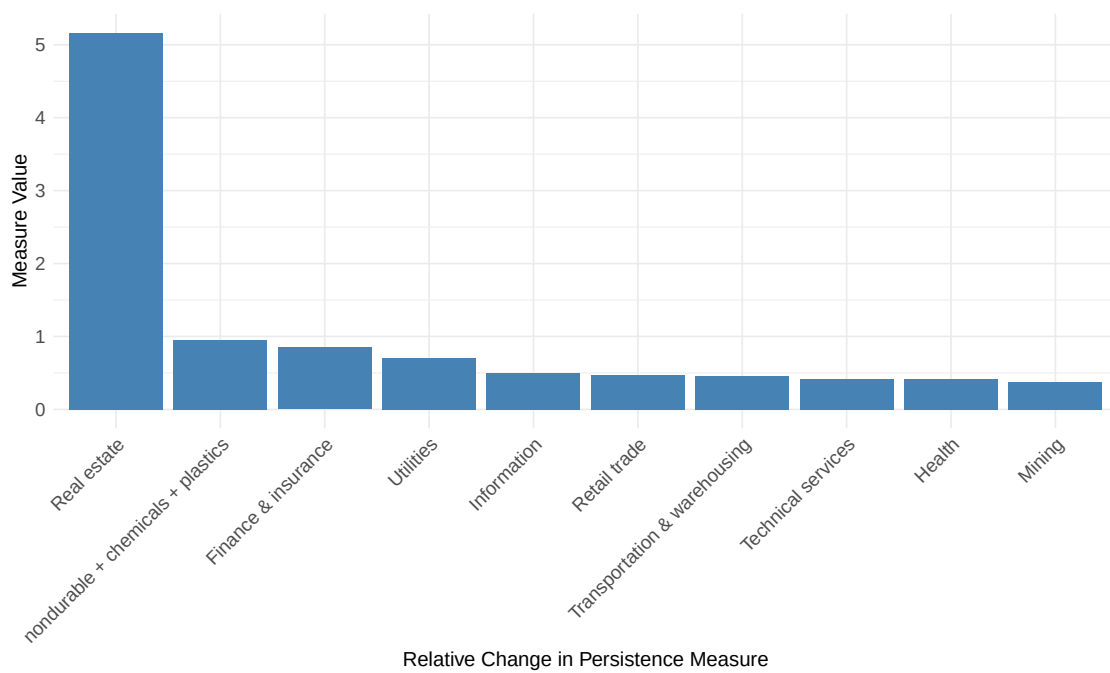


Figure 18: Identified top 10 bottleneck sectors from time-to-build, measured using distance ratio.